

Zombie Arcade

Official Game Document

Document Type: Post-Complete Game Design Overview (Version 1.0)

Project Status: Completed, October 2025

Elevator Pitch

'Zombie Arcade' is a top-down, retro arcade puzzle survival game inspired by the 1984 classic 'Oh Mummy'. Players navigate through grid-based container yards, surrounding containers with footsteps to reveal gameplay-altering chaos, deadly surprises, or the keys required to escape as they rack up Score to climb the Leaderboard.

Table of Contents

Elevator Pitch	1
Table of Contents	2
Game Concept	3
Game Summary	3
Genre & Target Audience	3
Target Platform	3
Unique Selling Points	4
Story & Setting	4
Premise	4
World & Environment	4
Characters	5
Narrative Flow	7
Gameplay Overview	7
Core Mechanics	7
Objectives	7
Rules & Systems	8
Progression	10
Level Design	11
Level Layout	11
Challenges & Puzzles	11
Rewards & Progression	11
Art & Visuals	12
Art Direction	12
Characters & Environment	13
HUD & UI Elements	14
Animations	18
Audio	20
Music	20
Sound Effects	20
Controls & User Interface	20
Controls	20
UI Flow	21
Technical Specifications	22
Engine & Tools	22
Platform Requirements	22
Performance Targets	22
Development & Testing Hardware	22
Primary Development Machine	22
Quality Assurance Testing	23
Postmortem	24
Future Plans	26
References & Inspirations	27

Game Concept

Game Summary

Zombie Arcade is heavily inspired by the 1984 classic retro arcade game *Oh Mummy* and is designed as a modern reinterpretation. The project aims to preserve the strategic grid-based gameplay of the original, while introducing modernized gaming mechanics and quality-of-life features to enhance the classic formula.

Set in the near future, players control an unnamed survivor during a zombie apocalypse, navigating endless container yards in search of supplies for survival. Each level consists of a grid of containers that can be opened by surrounding them with footsteps to reveal their contents. The contents include helpful or unhelpful **Power-Ups**, dangerous **Zombies**, **Score** and **Token** bonuses, **Keys** to escape the level, or nothing at all.

The game features two gameplay modes; **Retro**, which faithfully recreates the experience of *Oh Mummy* in the *Zombie Arcade* universe to offer a nostalgic arcade experience, and **Modern**, which expands on the foundations laid by Retro by introducing a variety of Power-Ups, a Token-purchasing upgrade system, a Dash ability for enhanced movement, Armour mechanics that create a deeper survivability system, and deadlier, self-upgrading Zombies.

The core objective is to survive for as long as possible to accumulate Tokens for upgrades (if in Modern mode) and Score for Leaderboard progression. Score can be accumulated in a variety of ways, from placing footsteps to opening containers to killing Zombies to clearing the entire level – and more.

Genre & Target Audience

Zombie Arcade is a top-down, arcade puzzle survival game designed for players who enjoy score-chasing gameplay loops, quick decision-making, and competitive Leaderboard systems. The experience is designed to appeal to both retro gaming enthusiasts and modern players seeking fast-paced and replayable arcade challenges.

Recommended age: **PEGI 12** – Mild violence and threat.

Target Platform

The game targets Windows PC only. Console support is currently beyond scope, and while mobile support is not yet offered, it may become a target for future updates.

Unique Selling Points

- **Retro Reimagined:** A modern adaptation of the classic arcade puzzle game *Oh Mummy* with expanded mechanics, modern enhancements, and a new setting.
- **Two Game Modes:** With a faithful retro rendition in **Retro** and a modernized retelling in **Modern**, players can experience two ways to play the game.
- **Risk-Reward Gameplay:** Players must decide when to leave levels early for survivability or stay for that bit of extra Score as they navigate unpredictable, high-risk situations.
- **Competitive Online Leaderboards:** Dedicated online Leaderboards for each game mode encourage replayability and competitive score chasing.

Story & Setting

Premise

In the near future, the world has been ravaged by a deadly zombie virus, collapsing civilization and leaving scattered survivors struggling to endure the apocalypse. Players take control of an unnamed male survivor wandering endless container yards in search of supplies for survival. The yards contain sealed containers that could hold valuable resources, useful equipment, or unwanted dangers.

Dangerous Zombies roam the container yards in search of their next meal, forcing the survivor to navigate deadly, high-risk environments in search of the keys needed to escape. The game contains light narrative context to focus primarily on delivering a fast-paced experience centred on survival, strategy, and score progression – similar to that of *Oh Mummy*.

World & Environment

The game takes place within large industrial container yards consisting of rows of containers arranged in grid patterns that form the basis of the game's puzzle mechanics – directly inspired by *Oh Mummy*, which takes place inside Egyptian pyramids containing rows of chests that form the game's puzzle mechanics.

Each level represents a new section of the container yard, filled with sealed containers that must be surrounded by footsteps to open. The setting emphasizes simplicity and clarity to ensure players can easily read the grid layout and react quickly to the changing gameplay.



Figure 1: The second level container yard in Modern mode.

Characters

Unnamed Male Survivor (Protagonist)

Navigates endless container yards in search of supplies for survival. Adept at killing Zombies with a Crossbow (learned over the years of the apocalypse) and can also utilize a quick Dash ability for a last-chance getaway (Modern mode only).

In Modern mode, the survivor can develop stronger abilities through Perma-Skills, which are permanent upgrades that can be bought using earned Tokens through gameplay.



Figure 2: The unnamed male survivor of the game.

Zombie Variants (Antagonists)

The enemies of the game are the handful of Zombie variants. There is a total of five different Zombies, each with their own unique behaviours and statistics, which will be displayed in a table below.

Zombie Variant	Behaviour	Threat Level
WalkerZ	The most basic variant with low statistics across the board. Easy to avoid and kill, primarily serves as a ‘filler’ enemy.	Low
DogZ	A unique ‘pack-spawning’, ‘timed’ variant. Can spawn solo or up to three before relentlessly chasing the player for a duration until they die. Low health and attack damage with high attack speed.	Medium
VomiterZ	Large tank-like Zombies with high health and detection range. Spits Vomit projectiles in the player’s direction which rest on the floor for a duration after.	Medium
RunnerZ	A faster variant of the WalkerZ, with the same low statistics except for very high speed, keeping the player on edge with expert movement to avoid.	High
ApexZ	The elite variant with very high attack damage, high health and detection range, and moderate speed. Avoid at all costs.	Extreme



Figure 3: WalkerZ.



Figure 4: DogZ.



Figure 5: VomiterZ.



Figure 6: RunnerZ.



Figure 7: ApexZ.

Narrative Flow

The game does not follow a structured narrative progression as it delivers a minimal story backdrop to support the main feature of the game – which is the gameplay. There is some progression throughout, as each level represents a new section of the container yards before a Checkpoint is reached, but no additional story elements are delivered.

Each run by the player represents another attempt by the survivor to explore the dangerous environments as they search for valuable resources while avoiding the ever-present undead threat.

Gameplay Overview

Core Mechanics

The gameplay revolves around navigating grid-based levels to reveal container contents, avoid Zombies, and escape once both Keys have been found. The player has access to the following mechanics:

- **Movement:** To traverse the level.
- **Dashing:** To ‘Dash’ forward in the last inputted direction – only available in Modern mode.
- **Shooting:** To Shoot arrows from a Crossbow in the last inputted direction.
- **Pause:** To pause the game and showcase the Pause Menu.

Objectives

The main objective is to survive while racking up Score to climb the Leaderboard. There are a variety of actions that can earn or lose Score, such as placing footsteps, killing Zombies, opening containers, clearing the entire level, Power-Up effects, and more. Players can choose to exit levels before full completion (all containers opened and all Zombies killed) if they have found both the two Keys – one Gold and one Silver.

The game indefinitely continues with difficulty gradually increasing as Zombie numbers grow. Additionally, in Modern mode, Zombies automatically upgrade their gameplay statistics (health, attack damage, detection range, etc) to match the player’s own upgrades.

Rules & Systems

Both game modes follow some core gameplay rules, such as:

- Containers must be fully surrounded with footsteps before their contents are revealed.
- Each level contains two Keys – one Gold and one Silver – that must be retrieved in order for the player to exit.
- After every five levels, a Checkpoint is reached where the number of Zombies per level is increased.
- Zombies continuously roam the level until they are either killed, the level ends, or the player enters their detection radius, in which they then pursue the player.
- If the player's Health reaches zero, their run ends.

Modern mode introduces additional systems such as Armour as well as Health, a Dash ability for advanced movement, a multitude more Power-Ups with different categories based on their behaviour, and permanent skill upgrades which the player can buy at each Checkpoint, while Zombies automatically upgrade at each Checkpoint. Below is a table explaining all Power-Ups in *Zombie Arcade*.

Power-Up Name	Power-Up Effect	Power-Up Category	Power-Up Rarity	Game Mode Availability
Armour Vest	Restores a portion of Armour	Positive	Common	Modern
Bonus Stack	Adds a stack of additional Score and Tokens to the Player's current Score and Tokens total	Positive	Common	Retro, Modern
Energy Drink	Gives a movement speed buff for a duration	Positive	Common	Modern
Med Kit	Restores a portion of Health	Positive	Common	Modern
Double Or Nothing	Doubles all Score and Tokens earned for a duration	Positive	Rare	Modern
Nice Shoes!	Grants immunity to all Vomit attacks for a duration	Positive	Rare	Modern
Ninja	Restores a Dash	Positive	Rare	Modern
Key Radar	Reveals the location of one of the Containers holding one of the Keys	Positive	Very Rare	Modern
The Somewhat Cure	Cures a random Zombie in the Level, removing their aggro towards the Player while helping place footsteps to open	Positive	Very Rare	Modern

	Containers for a duration until they are killed			
Booby Trap	Gives a movement speed nerf for a duration	Negative	Common	Modern
Fatigue	Removes the use of the Dash ability for a duration	Negative	Common	Modern
Starvation	Starves the Player for a duration, dealing frequent low tick damage	Negative	Common	Modern
Wrong Turn	Spawns a low-medium threat Zombie (WalkerZ, VomiterZ)	Negative	Common	Retro, Modern
DiZaster	Spawns a high-extreme threat Zombie (ApexZ, RunnerZ)	Negative	Rare	Retro, Modern
Who Let The DogZ Out?	Spawns a random number (in a range) of DogZ	Negative	Rare	Retro, Modern
Alarm Clock	Alerts all Zombies to the Player's location for a duration	Negative	Very Rare	Modern
No Luck	Removes any and all currently active Positive Power-Up effects	Negative	Very Rare	Modern
Food	Removes the Starvation Negative Power-Up effect	Special	Common	Modern
Crossbow	Allows the Player to shoot arrows at – and deal damage to – Zombies	Special	Rare	Retro, Modern
Guts, Guts, Guts!	Grants immunity to all Zombies' aggro for a duration	Special	Rare	Modern
Why So Serious?	Closes any and all open Containers and shuffles their positions across the Level, and additionally removes any and all placed footsteps in the Level	Special	Very Rare	Modern
Perma-Skill	Grants one free-of-charge Perma-Skill upgrade from a random choice of three	Special	Super Rare	Modern

All Power-Ups that are available in Retro (Bonus Stack, Wrong Turn, DiZaster, Who Let The DogZ Out?, and Crossbow) are all based on the Power-Ups you can earn in *Oh Mummy*. In that game, players can unlock the following from chests: nothing (empty), a Money Rune, a Pharaoh's Sarcophagus, a Key, a Scroll, or an enemy Mummy. The Money

Rune gives players bonus score, just like *Zombie Arcade*'s Bonus Stack, while the Pharaoh's Sarcophagus and Key act as the level keys needed to exit while *Zombie Arcade* contains two different coloured keys. The Scroll allows players to defeat an enemy Mummy without taking damage, while *Zombie Arcade* utilizes a Crossbow, and the enemy Mummy spawners in *Oh Mummy* are shown as three separate Zombie spawners (Wrong Turn, DiZaster, Who Let The DogZ Out?) for the different Zombie variants of *Zombie Arcade*.

Progression

The player faces tougher runs the further they advance due to the Zombie increases at each Checkpoint. The main aim for players is to earn as much Score as possible for higher positions on the Leaderboard, encouraging them to stick around in the levels longer. Scores are submitted to online Leaderboards for players to compare performance globally. Additionally, in Modern mode, players earn Tokens to spend at the Perma-Skill Shop for permanent upgrades to their survivor, helping them survive longer in the runs for higher scores. Below is a table explaining each of the Perma-Skills' effects.

Perma-Skill Name	Perma-Skill Effect
Army Veteran	Increases max Armour
Black Belt	Increases max Dashes
Bodybuilder	Increases max Health
Eight Ball	Increases chance of finding Positive Power-Ups
Fighter	Increases Damage Mitigation
Lucky-Lucky	Increases chance of finding Special Power-Ups
Scavenger	Decreases chance of finding empty containers
Speedster	Increases movement speed

Level Design

Level Layout

Levels are structured exactly like that of the levels from *Oh Mummy*, in a grid-like formation of five containers in four rows, equalling a total of 20 containers. This layout promotes predictable gameplay, which is where the dangerous Zombie variants and unpredictable container contents come in to mix up the gameplay to promote high-risk, high-reward strategic pathing and decision making which rewards fast reactions.

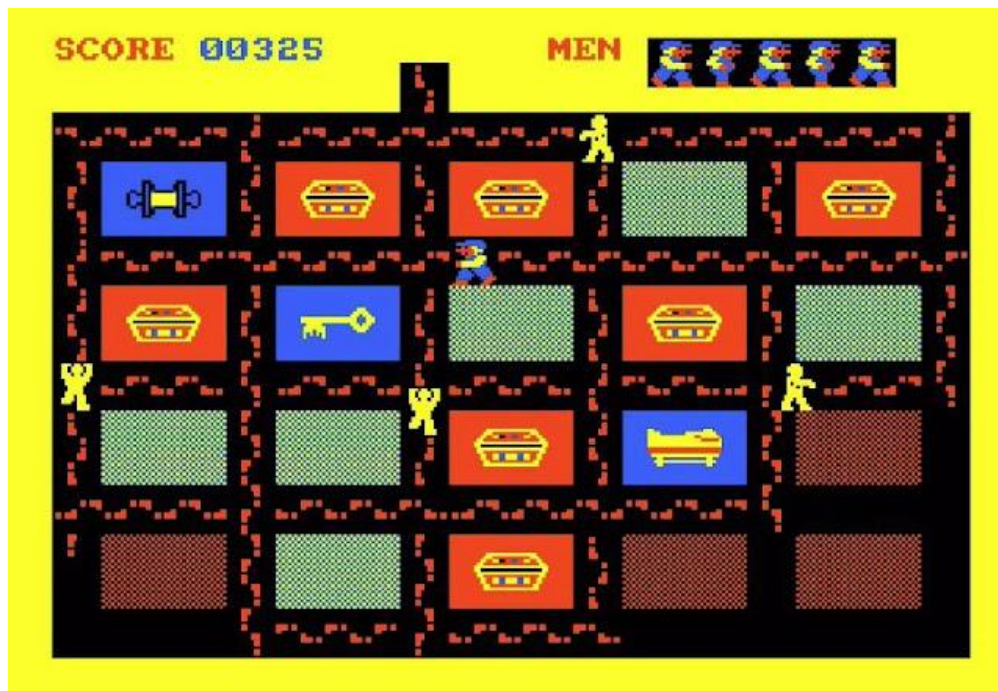


Figure 8: The level layout of *Oh Mummy*, which inspired *Zombie Arcade*.

Challenges & Puzzles

The main challenge of the game lies in balancing exploration with survival. Exploring until the two Keys needed to escape are found is a necessity, but exploring beyond that to earn additional Score for that push up the Leaderboard comes as a risk players need to be wary of. Revealing additional containers will earn more Score and can even contain powerful advantages, but they can also reveal more Zombies to deal with or negative Power-Ups that complicate the situation.

Rewards & Progression

Players consistently earn Score and Tokens (the latter only in Modern mode) while they play the game which contributes to higher Leaderboard placements and permanent upgrades for better runs. The longer the player survives, the more Score and Tokens they earn, but also the greater the threat of death will be.

Art & Visuals

Art Direction

The game uses classic retro 2D pixel art inspired by *Oh Mummy* and the many other classic arcade games of the 1980's. The visual design intentionally emphasizes clarity and readability to support the fast-paced nature of the gameplay while remaining faithful to retro arcade video games. Simple environmental details, distinct enemy sprites, and clear colour-coded container states help players quickly identify important elements of the gameplay, all while appealing to the nostalgia of retro arcade games.

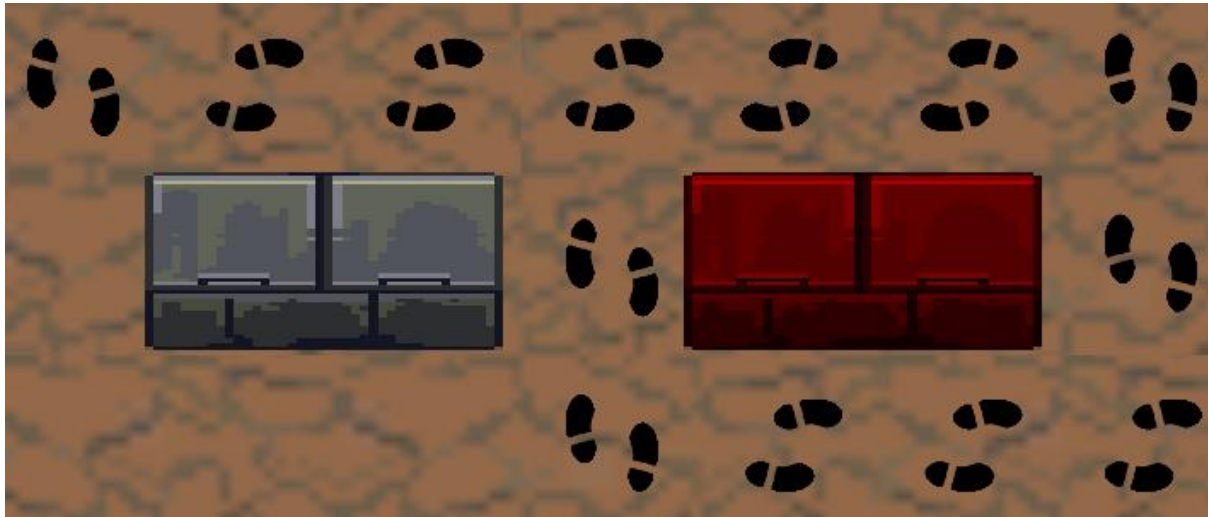


Figure 9: The two container states. Red indicates an open container, while grey (no sprite colour overload) indicates an unopen container.

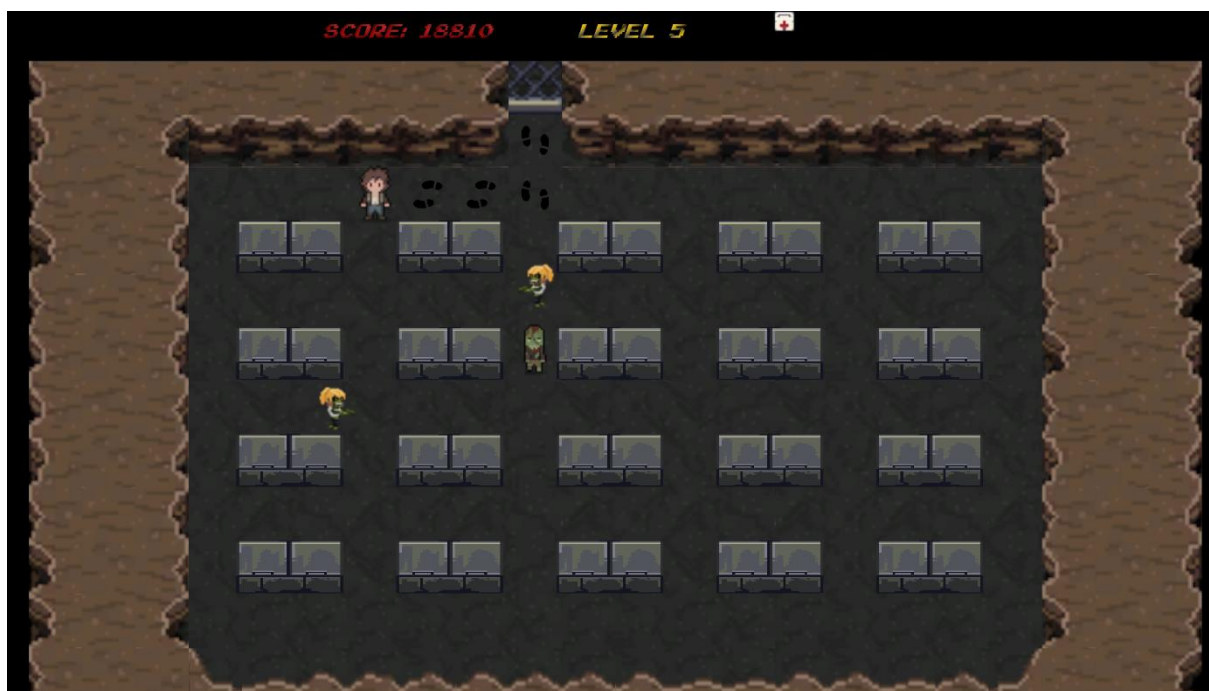


Figure 10: The art of *Zombie Arcade*. The level shows the rundown atmosphere of the world, with aging landscapes and deteriorating containers.

Characters & Environment

The player character is represented as a simple pixel-art sprite with a rugged, rundown look to show how the years of apocalypse survival has affected them. This goes for the Zombies, too, who each have a distinct characteristic look to help players identify their variant.

The environment is very simplistic, featuring worn-and-torn landscapes and footpaths, along with aging containers. Finally, the Power-Ups of the game each have a distinct apocalyptic theme to their sprite which matches the story and aesthetic of the game, while also clearly separating themselves from other Power-Ups to allow the player to learn them.



Figure 11: Concept art for an apocalyptic world that inspired the theme and aesthetic of Zombie Arcade. Image created by Colton Force from ArtStation.



Figure 12: Concept art for a container yard that inspired the theme and aesthetic of *Zombie Arcade*. Image created by Maarten Hof from ArtStation.

HUD & UI Elements

The HUD is vital for gameplay as it features numerous important elements, such as:

- **Current Level Indicator:** Displays the level the player is currently on.
- **Current Armour Indicator:** Displays the player's current Armour (Modern mode only).
- **Current Health Indicator:** Displays the player's current Health.
- **Current Dashes Indicator:** Displays the player's current available Dashes (Modern mode only).
- **Current Tokens Count:** Displays the player's current Tokens value (Modern mode only).
- **Current Score Count:** Displays the player's current Score value.
- **Currently Active Power-Ups Panel:** Displays all the currently active Power-Ups with a small descriptive text of their effect, and a border-timer to showcase its duration.



Figure 13: The HUD of Zombie Arcade in Retro mode, displaying the current level, Score value, Health value, and any currently collected items and/or active Power-Ups.



Figure 14: The HUD of Zombie Arcade in Modern mode, displaying all of the above from Retro, as well as Tokens value, and Armour and Dashes value.

The Main Menu is centralized on a retro synthwave backdrop to showcase a classic look, featuring the following options:

- **Game Title:** Big and bold.
- **Play Button:** Displays the Game Mode Selection Menu before starting the game.

- **How To Play Button:** For viewing the rules of the game, along with additional helpful information.
- **Options Button:** For viewing and editing the controls scheme or editing the audio volume.
- **Credits Button:** For viewing the credits of the game.
- **Quit Button:** For quitting the game.

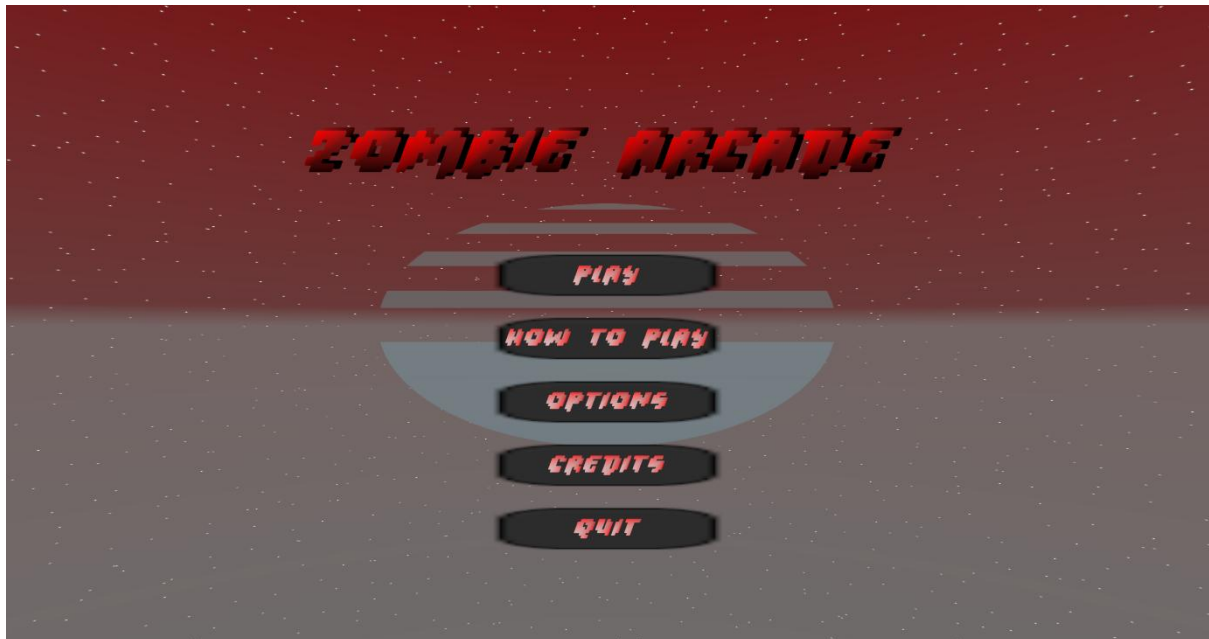


Figure 15: The Main Menu.

The Pause Menu appears as a small box overlapping the paused state of the main game, displaying the following options:

- **Resume Button:** For resuming the game.
- **How To Play Button:** Same function as the Main Menu.
- **Options Button:** Same function as the Main Menu.
- **Quit Button:** Quits to the Main Menu.



Figure 16: The in-game Pause Menu.

The Checkpoint Menu follows a similar synthwave aesthetic as the Main Menu, and displays the player's current Score (and Tokens if in Modern mode), while featuring a couple options, with an additional option if in Modern mode:

- **Next Level Button:** Starts the next level, with all upgrades and increases being applied.
- **Shop Button:** Takes the player to the Perma-Skill Shop, only available in Modern mode.
- **Quit Button:** Quits to the Main Menu.



Figure 17: The Checkpoint Menu.

The Perma-Skill Shop Menu also uses the synthwave aesthetic but with a more cool, nighttime vibe to alter its personality. It features a variety of buttons for purchasing multiple Perma-Skills, along with a button for starting the next level like that shown in the Checkpoint Menu.



Figure 18: The Perma-Skill Shop Menu.

Animations

Player

The player uses movement animations for top-down walking left and right. Additionally, death and shooting animations are used for when the player's health is zero or when they have – and shoot – the Crossbow Special Power-Up. The Dash ability is represented as a quick 'teleport', so no animation is required, while damage is represented as a flash of red over the character sprite instead of an animation.

Zombies

The Zombies also use movement animations like the player, as well as death animations for when they are killed. Their damage is also represented as a flash of red over the sprite instead of a dedicated animation. Unique to the Zombies is their attack animations, where each of the five have their own attack animations for when they are within attack range of the player.

Environment

There are no environmental animations, with the closest thing being the containers, which change to a red sprite when opened, or a green sprite when they contain one of the escape Keys and are pinged by the Key Radar Positive Power-Up.

Other

While the player uses a shooting animation for the Crossbow, an animation for the arrow to travel is also used to bring the mechanic together. Additionally, while the Vomiter Zombie has its own attack animation, it is paired with animated Vomit projectiles to bring its attack to life. An additional animation is used for the Vomit projectiles that splash on the floor.

Finally, there is a HUD animation for active Power-Ups, where any currently active show an animated border timer around its sprite to showcase how long its duration has remaining.

Audio

Music

Retro-inspired, 8-beat music tracks are used throughout *Zombie Arcade* to further bring its nostalgic, arcade aesthetic together. The soundtrack was precisely chosen to maintain an energetic and catchy tempo to reinforce the fast-paced gameplay and score-chasing atmosphere.

Sound Effects

Sound effects in the game play a pivotal role. *Zombie Arcade* is a high-risk, high-reward, fast-paced game where multiple events can be happening at once. While the HUD has been designed to showcase helpful information to the player, they may not always be able to look due to the dangerous and alerting gameplay. Therefore, the sound effects have been carefully selected to deliver distinct audio cues related to important updates, such as Power-Up activations or damage taken.

Every Power-Up in the game has its own unique audio cue for the player to learn when they uncover it, allowing them to learn each Power-Up by its audio alongside its sprite. The player has a sound effect for taking damage and shooting arrows, while Zombies also have sound effects for taking damage and being killed. UI elements, such as button clicks, also utilize appropriate audio cues to deliver a quality user experience.

Controls & User Interface

Controls

- **Movement (WASD/Arrow Keys):** Allows the player to traverse the game world.
- **Dash (Left Shift):** Allows the player to Dash in their last inputted direction for quick traversal.
- **Shoot (Spacebar):** Allows the player to shoot arrows from the Crossbow Special Power-Up in their last inputted direction to deal damage to – and kill – Zombies.
- **Pause/Menu (Esc):** Allows the player to pause the game, displaying the Pause Menu.

Players can swap the control binds for Dash, Shoot, and Pause/Menu inside the Options Menu, allowing them to configure the best setup tailored to their preference. With *Zombie Arcade* being a fast-paced and competitive game, it felt essential to include a key rebind system.

UI Flow

The HUD can contain much information at once, due to the informative environment of the game. The layout has been organized in the neatest, clearest way possible for players to navigate. All menus and HUD elements share consistent colour palettes and background themes to ensure visual synchronization tailored to a retro synthwave aesthetic. Menus support full mouse navigation for ease-of-use.

Technical Specifications

Engine & Tools

Zombie Arcade was developed with Unity v6, with C# for scripting. Rider (coding software) and Audacity (audio manipulation software) was also used in the creation of the game.

Platform Requirements

	Minimum	Recommended
OS	Windows 10 64-bit	Windows 10/11 64-bit
Processor	AMD FX-3600 / Intel Core i3-6100	AMD Ryzen 5 2600 / Intel Core i5-8400
Memory	4 GB RAM	8 GB RAM
Graphics	NVIDIA GeForce GTX 750 Ti / AMD Radeon R7 260X	NVIDIA GeForce GTX 1060 / AMD Radeon RX 580
VRAM	2 GB	4 GB
Storage	1 GB available space	2 GB available space (SSD)
DirectX	Version 11	Version 11

Performance Targets

- **Frame Rate:** 60 FPS at 1080p on Minimum Specs, 120 FPS at 1080p on Recommended Specs
- **Resolution:** 1920x1080
- **Loading Times:** ~1-2 seconds on SSD; ~3-6 seconds on HDD

Development & Testing Hardware

Primary Development Machine

OS	Windows 11 64-bit
Processor	AMD Ryzen 7 7800X3D
Memory	32 GB RAM
Graphics	NVIDIA GeForce RTX 4070 SUPER
VRAM	12 GB
DirectX	Version 12

Quality Assurance Testing

The game underwent multiple rounds of internal playtesting to ensure stable performance and properly working intended game design mechanics, including balanced gameplay. Testing focused on verifying the behaviours of the Zombie variants, the probability distribution of container contents, and the scaling difficulty across Checkpoints, amongst others. Performance tests were conducted across multiple PC configurations by multiple candidates.

Postmortem

What Went Well

- Successfully recreated the core gameplay loop of *Oh Mummy* while expanding on it with modernized features.
- The creation of the two games modes (Retro and Modern) offers great replayability and varying gameplay which appeals to a wider audience. Retro delivers a faithful recreation of the classic formula, while Modern delivers an expanded experience with additional gameplay mechanics, Power-Ups, and upgrades.
- Visual design of the game creates a beautifully retro apocalyptic world to set the game's theme apart from that of *Oh Mummy*.
- Multiple AI behaviours were introduced for varying enemies, along with a smart automatic upgrade system.
- Playtesting was thoroughly conducted at perfect times to document and alter specific gameplay mechanics early before becoming too big an issue later on.
- With this being my fourth official game project, code has improved significantly as a wide range of new techniques were learned and implemented, such as mathematical probability systems for Power-Up category chance (which can be altered through purchased Perma-Skills), while the Perma-Skill Special Power-Up smartly detects the lowest tier upgrades the player has purchased to showcase in the Power-Up – used to encourage more balanced character progression and prevents players from repeatedly stacking the same upgrades.
- Several progression systems were also implemented using efficient and scalable code structures using helper functions and parent-child systems. Player and Zombie upgrades utilize these systems.

What Could Have Been Better

- Early design documentation could have been more structured to streamline later stages of development. Documentation was created but heavily relied on notes and ideas that were not professionally published for expandability.
- Additional, powerful setup tools – such as Trello and HacknPlan – could have been used to help early design documentation and to ensure focus was aimed at the right areas throughout the entire development process.

Key Lessons Learned

- Designing around simple but well-defined gameplay loops can produce highly replayable and expandable experiences.
- Structured documentation helps streamline development and prevents unnecessary redesigns later in production.

- Frequent playtesting is essential for identifying gameplay balance issues early, ensuring the final experience remains fair and fun.
- Using scalable and efficiently designed code, including helper functions, modular systems, and inheritance structures, makes it easier to expand progression systems without rewriting large portions of code. These approaches significantly improved development efficiency and maintainability.
- Designing container contents around probability over fixed outcomes proved effective for maintaining replayability. Creating upgrades for the player to purchase to alter these probabilities creates a dynamic risk-reward system which allows players to feel that their decisions having a meaningful impact on the gameplay. Additionally, while randomness is a welcome addition to these container contents, too much unpredictability can make gameplay feel unfair. Creating Power-Up rarities (Common, Rare, Very Rare, Super Rare) helps maintain randomness without undermining player skill or strategic decision making.

Future Plans

Zombie Arcade is developed as a solo standalone project, with no ideas or intentions of further sequel development. It was always planned as a small experience based on *Oh Mummy*, delivering some classic retro arcade action in the modern age. However, throughout development, plenty of ideas came to mind about potential downloadable content (DLC) packages. These would include a variety of new Zombies – and maybe even some human enemies – more Power-Ups and Perma-Skills, new abilities for the player, and some new, gameplay altering mechanics such as nighttime map variants where the player is limited to a small circle of light instead of seeing the entire level.

References & Inspirations

Historical

The game's mechanics are completely built on grid-based movement – just like *Oh Mummy*. The 1984 classic video game serves as a major inspiration for the project in almost every way (mainly core gameplay mechanics), but it was not the first video game to utilize grid-based movement. '*Pac-Man*' is arguably the earliest, most famous example of the movement, while '*Amidar*' is seen as *Oh Mummy*'s inspiration, making it inspiration for *Zombie Arcade*, too.

Games

Aside from the obvious inspiration in *Oh Mummy*, *Zombie Arcade* also took small inspiration from a variety of other retro arcade games that include score-chasing, Power-Up altering gameplay with automatic difficulty scaling. These games include the *Pac-Man* series, '*1942*', and '*Space Invaders*'.

Other

There are plenty of modern reimagining's of retro arcade games that mix the classic formula with modernized gameplay – and these all have served as inspiration for this project. Studying how those games successfully stay truthful to its source material while expanding with its unique modern systems helped in the creation of the game. The games I mentioned above (*Pac-Man*, *1942*, *Space Invaders*) have all seen modern renditions, as well as a plethora of other games.

Additionally, the theme of apocalypse and zombies served as a major reference for the game's world, story and setting. To set *Zombie Arcade* apart from its *Oh Mummy* inspiration, the best aspect of the game to change was the setting and theme, and for this I turned to apocalypses (specifically zombie apocalypses), as they offer a variety of items to be used for Power-Ups, as well as much freedom in the zombie creation department with varieties.